

Radiologist Interaction with Artificial Intelligence-Generated Preliminary Reports: A Longitudinal Multireader Study

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Abstract

Objectives: To investigate the integration of multimodal AI-generated reports into radiology workflow over time, focusing on their impact on efficiency, acceptability, and report quality.

Methods: A multicase, multireader study involved 756 publicly available chest radiographs interpreted by five radiologists using preliminary reports generated by a radiology-specific multimodal AI model, divided into 7 sequential batches of 108 radiographs each. Two thoracic radiologists assessed the final reports using RADPEER criteria for agreement and 5-point Likert scale for quality. Reading times, rate of acceptance without modification, agreement, and quality scores were measured, with statistical analyses evaluating trends across seven sequential batches.

Results: Radiologists' reading times for chest radiographs decreased from 25.8 seconds in batch 1 to 19.3 seconds in batch 7 ($P < .001$). Acceptability increased from 54.6% to 60.2% ($P < .001$), with normal chest radiographs demonstrating high rates (68.9%) compared with abnormal chest radiographs (52.6%; $P < .001$). Median agreement and quality scores remained stable for normal chest radiographs but varied significantly for abnormal chest radiographs (all $P < .05$).

Discussion: The introduction of AI-generated reports improved efficiency of chest radiograph interpretation, and acceptability increased over time. However, agreement and quality scores showed variability, particularly in abnormal cases, emphasizing the need for oversight in the interpretation of complex chest radiographs.

Key Words: Chest radiographs, generative AI, radiologist-AI interaction, report generation

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INTRODUCTION

Artificial intelligence (AI) is increasingly used in radiology to enhance diagnostic accuracy and efficiency [1,2]. Initial efforts primarily focused on classification tasks, such as detecting pulmonary nodules or triaging abnormal examinations, but

recent developments extend beyond single-task detection or classification. Among these, multimodal generative AI, which can integrate image analysis and natural language generation, shows promise for automating radiologic reporting by producing preliminary AI-generated reports [3,4]. Although this technology offers significant potential, its clinical validation is essential to determine its effects on report quality, radiologists' efficiency, and patient safety before widespread implementation.

The integration of AI into radiology has also raised concerns about the possibility of radiologists depending too heavily on AI-generated suggestions, which could influence decision making and diagnostic outcomes [5]. Although automation bias and related phenomena have been described in prior studies [6,7], their implications in radiology, particularly in the setting of generative AI, remain incompletely understood. In radiology, the potential impact of generative AI warrants careful consideration, as these systems produce complete narrative

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Table 1. Patient demographic and CXR label characteristics for the reader study dataset

Characteristics (n = 756)	Values
Age (median)	62
Range	2-97
Gender (M/F)	403/353
Projection, n (%)	
PA	582 (77.0)
AP	174 (23.0)
Normal CXRs	
n	223
Age (median)	52
Gender (M/F)	84/139
Abnormal CXRs	
n	533
Age (median)	68
Gender (M/F)	319/214

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reports rather than isolated cues or annotations. Such AI-generated preliminary reports may enhance efficiency, yet draft that contains inaccuracies could influence radiologists' interpretations, potentially introducing diagnostic variability or reducing independent verification. Careful evaluation of both advantages and risks is necessary to determine how AI-generated reports can be integrated into clinical workflows in a way that supports diagnostic accuracy and maintains radiologist oversight.

Therefore, the current study investigates how radiologists interact with AI-generated reports in a multireader, multicase study setting, focusing on the impact of AI assistance on interpretation time, decision-making processes, and report quality with repeated exposure.

METHODS

This retrospective study using public dataset was exempted from review by the local institutional review board. The study used 756 frontal chest radiographs from the publicly available PadChest dataset, randomly selected via Python (3.11.5) [8]. Out of these, 223 (29.5%) were labeled as normal (without any abnormality), and 533 (70.5%) were labeled as abnormal (with abnormalities) based on annotations provided by the original publisher (Table 1). The study used a radiology-specific multimodal generative AI model for chest radiograph report generation (KARA-CXR, Kakaobrain, Seoul, South Korea, for research use only). The

model processes chest radiographs as input and generates corresponding radiologic reports as output. It identifies abnormalities using an encoder-classifier system and produces reports via a transformer-based captioner, designed specifically to produce "findings-only" radiology reports, without additional impressions or recommendations [4].

Five radiologists, with 7 to 15 years of experience, including three specializing in chest, one in breast, and one in general radiology, independently interpreted the chest radiographs by reviewing and directly editing the AI-generated preliminary reports to create the final version, while being blinded to clinical and label information [9]. Chest radiograph interpretation was conducted using a custom-built reporting platform. Before the study, each radiologist reviewed 10 sample chest radiographs (not included in the study cohort) to become familiar with the reporting environment. The 756 study radiographs were divided into 7 sequential batches of 108 cases and interpreted over a 1-month period in November 2023; no fixed time interval was enforced between batches. Case order was randomized, and reading times were recorded automatically by the system.

Thereafter, two experienced thoracic radiologists with over 20 years of experience, blinded to clinical information and reader experience, assessed the agreement and quality of the final reports. Agreement scores were rated using RAD-PEER scoring system as follows: (1) clinically significant discrepancy; (2) clinically nonsignificant discrepancy; (3) clinically significant understandable miss; (4) clinically nonsignificant and understandable miss, which is justifiable given the complexity, subtlety, or difficulty of the case; and (5) complete agreement [10]. Quality scores of the final reports were assessed subjectively on a 5-point Likert scale: (1) very poor; (2) poor; (3) average; (4) good; and (5) excellent. Figure 1 illustrates the study design and workflow, and Figure 2 demonstrates examples of chest radiographs and the corresponding AI-generated reports.

For each batch, average reading times and the acceptance rate of AI-generated reports were analyzed. Acceptability was defined as a case in which no revisions were made to the AI-generated report before it was approved by the radiologist. In addition, we assessed the proportion of cases with complete agreement, median agreement, and quality scores. Complete agreement was defined as cases when two experienced thoracic radiologists, in consensus, rated the final report as 5, complete agreement.

Statistical comparisons were made between radiographs with and without abnormal findings, as well as between thoracic and nonthoracic radiologists. Variables that demonstrated normal distribution were analyzed using *t* tests. Nonnormally distributed and ordinal data (eg, agreement and quality scores) were assessed using the Mann-Whitney *U* test for the comparison between two groups. Additionally,

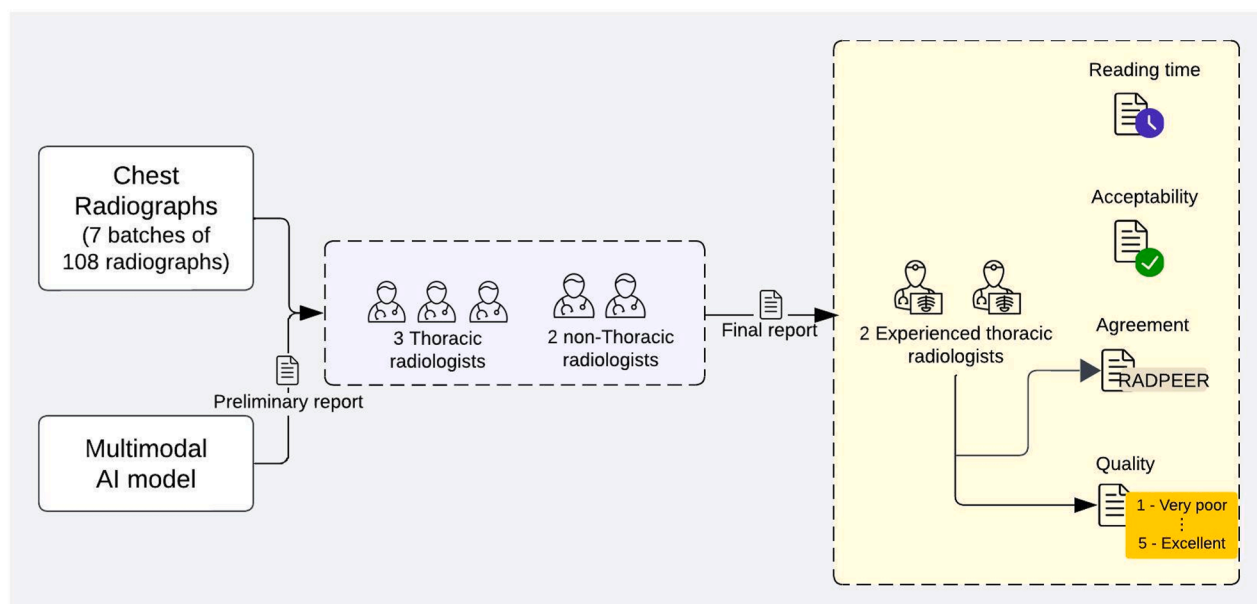


Fig. 1. Study design and study workflow, starting with chest radiographs (7 batches of 108 radiographs each) processed by the multimodal artificial intelligence (AI) model to produce preliminary reports. The chest radiographs and AI-generated reports were presented to three thoracic radiologists and two nonthoracic radiologists for interpretation. Reading time, acceptability, and agreement (using RADPEER criteria), quality ratings (on a scale of 1 = very poor to 5 = excellent) assessed by two experienced thoracic radiologists on the final reports are analyzed.

batch analysis was conducted to assess changes in reading time, acceptability, agreement, and quality scores over the course of the study. Linear regression was used to analyze trends in reading times. The Cochran-Armitage test for trend was applied to evaluate changes in acceptability and complete agreement across batches. Ordinal logistic regression was used to assess trends in agreement and quality scores, with batch number as the independent variable in all models.

Python's extensive libraries (Python Software Foundation, Wilmington, Delaware), such as Pandas, SciPy, and statsmodels, were used for statistical testing. For all tests, $P < .05$ was considered indicative of a statistically significant difference.

RESULTS

Reading times for chest radiographs significantly decreased over batches, from 25.8 ± 29.7 seconds in batch 1 to 19.3 ± 29.0 seconds in batch 7 ($P < .001$). Reading times for normal radiographs were consistently shorter than those for abnormal radiographs in all batches ($P < .001$) (Fig. 3a).

Acceptability rates of AI-generated reports increased over batches, from 54.6% (295 of 540) in batch 1 to 60.2% (325 of 540) in batch 7 ($P < .001$) (Fig. 3b). Normal radiographs had higher acceptability rates (68.9% [768 of 1,115]) than abnormal radiographs (52.6% [1,403 of 2,665]) ($P < .001$). Interestingly, thoracic radiologists exhibited higher acceptance rates for AI-generated reports

(76.7% [1,739 of 2,268]) compared with nonthoracic radiologists (28.6% [432 of 1,512]) ($P < .001$).

Complete agreement rates of final reports rated by two experienced thoracic radiologists did not change significantly across batches, 52.2% (282 of 540) in batch 1 to 63.2% (343 of 540) in batch 7 ($P = .087$) (Fig. 3c). Complete agreement was achieved in 56.6% (2,139 of 3,780) of all cases, with normal radiographs showing a higher agreement rate (57.7% [643 of 1,115]) compared with abnormal radiographs (56.1% [1,496 of 2,665]) ($P < .001$).

Agreement scores of the final reports had a consistent median of 5.0 (interquartile range [IQR] 4.5-5.0) across all batches and groups, showing no statistically significant change across batches for normal radiographs ($P = .570$). In contrast, abnormal radiographs exhibited statistically significant differences across batches ($P = .003$). Quality scores maintained a median of 4.5 (IQR 4.5-5.0) across batches, with normal radiographs generally receiving higher scores than abnormal radiographs ($P < .001$). Additionally, quality scores demonstrated statistically significant differences across batches for both normal and abnormal radiographs ($P < .001$ for both).

Table 2 displays specific values for reading times, acceptability, proportion of complete agreement, agreement, and quality scores of the final reports across radiologist groups, normal and abnormal radiographs, and batches, with corresponding P values for each comparison.

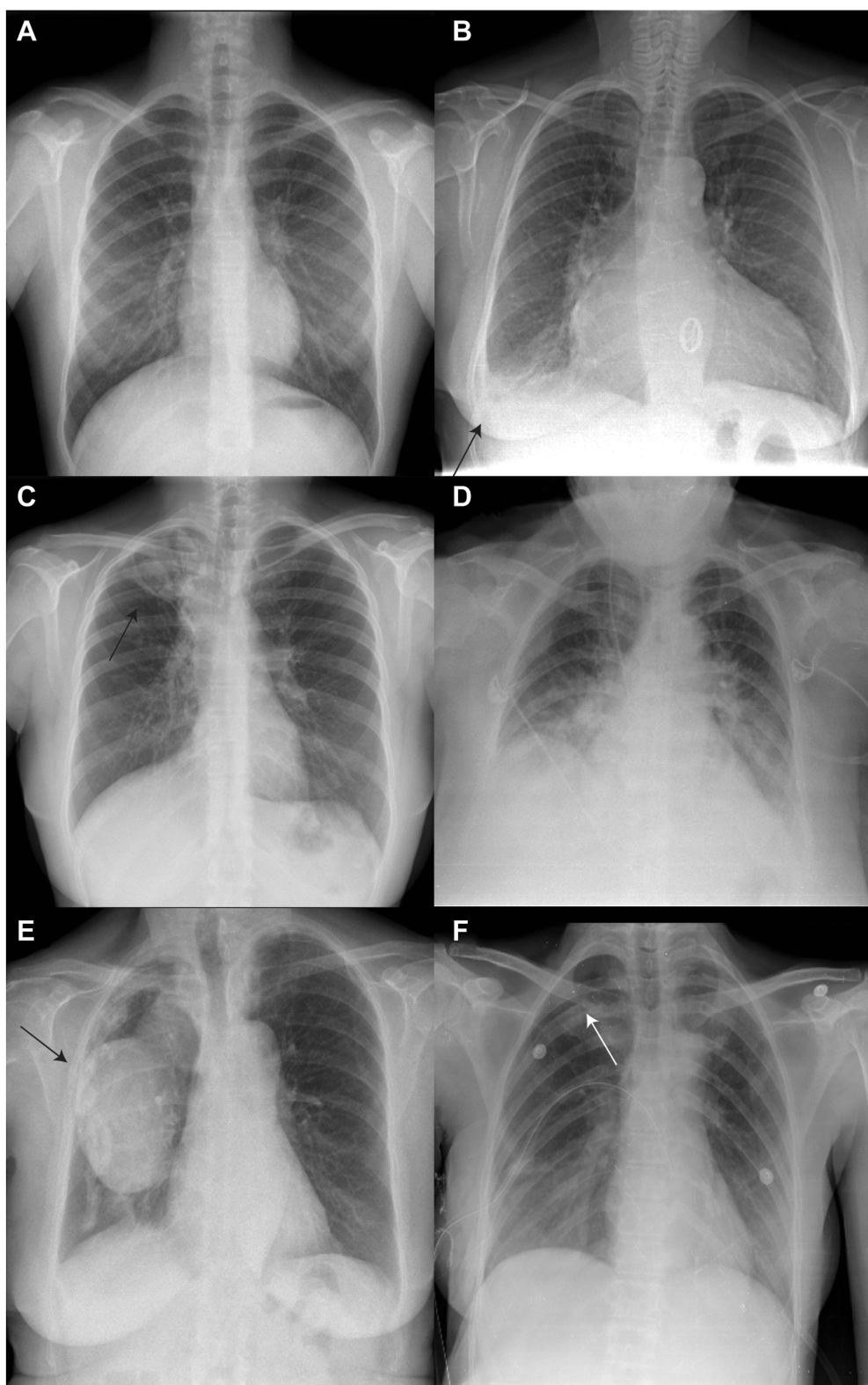


Figure 2. Examples of multimodal artificial intelligence (AI)-generated preliminary reports. (a) An example of a radiograph with an AI-generated report that all five radiologists accepted without modification in the first batch. The AI-generated report mentions “Lungs clear. No pleural effusion. No cardiomegaly. No fractures of suspicious lesions.” (b) An example of a radiograph with an AI-generated report that all five radiologists accepted without modification in the last batch. The AI-generated report describes “S/P median sternotomy and cardiac valve replacement. Cardiomegaly. Small Rt. pleural effusion

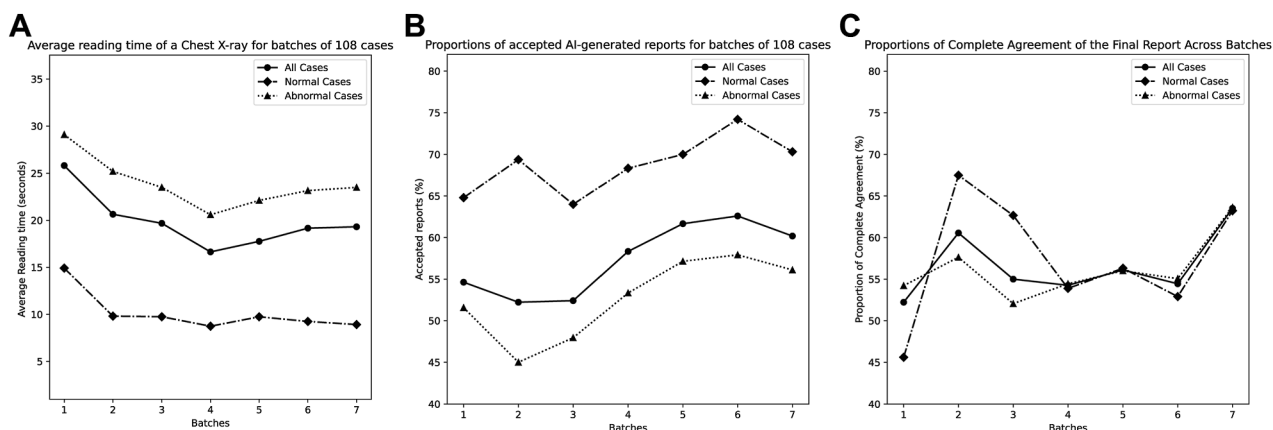


Fig. 3. Changes in reading time, acceptability, and complete agreement for artificial intelligence (AI)-generated chest radiograph reports across 7 batches of 108 cases. (a) Average reading time in seconds plotted for all cases (round marker), normal cases (diamond marker), and abnormal cases (triangle marker). (b) Proportion of accepted AI-generated reports in percentages displayed for all cases (round marker), normal cases (diamond marker), and abnormal cases (triangle marker). (c) Proportion of complete agreements of the final reports in percentages displayed for all cases (round marker), normal cases (diamond marker), and abnormal cases (triangle marker).

DISCUSSION

The results of this study demonstrate that AI-generated reports were associated with a reduction in chest radiograph interpretation times over the study period, with reading times decreasing over time from 25.8 ± 29.7 seconds in the first batch to 19.3 ± 29.0 seconds in the last batch ($P < .001$). Acceptability of AI-generated reports without modification increased from 54.6% in the first batch to 60.2% in the last batch ($P < .001$), suggesting growing confidence in AI-generated reports over time. Radiographs without abnormal findings had higher acceptability (68.9%) than those with abnormal findings (52.6%) ($P < .001$), reflecting the requirement of human oversight and revision in radiographs with abnormal findings. However, the proportion of complete agreement, rated by experienced thoracic radiologists, showed no significant change across batches ($P = .087$). Agreement scores for normal radiographs did not change over time ($P = .570$), and abnormal radiographs showed

variability across batches without discernable trend ($P = .003$). Similarly, quality scores for normal radiographs were consistent, and those for abnormal radiographs demonstrated significant variability without a clear trend ($P < .001$). Although the methodology is cited from our earlier publication [9], this study introduces a novel longitudinal analysis that examines radiologist-AI interaction over sequential batches, characterizing temporal variation in efficiency, acceptability, and report quality.

The current study provides an initial exploration of radiologist-AI interaction in radiology by employing multimodal generative AI for creating preliminary radiology reports. Unlike prior research that primarily addressed binary classifications or abnormality detection, this study investigates how AI-generated outputs may fit into radiologist workflows as preliminary reports [6]. By evaluating objective metrics such as reading times and subjective assessments like agreement and quality scores, our work

(black arrow). Patchy Rt. basilar opacity, possibly atelectasis or pneumonia. Mild pulmonary vascular congestion." (c) An example of a radiograph with an AI-generated report that only two of five radiologists accepted in the first batch. The AI-generated report states "Normal heart size. No pleural effusion. Volume loss and consolidation in the right upper lobe (black arrow). Left lung clear. No pneumothorax." (d) An example of a radiograph with an AI-generated report that four of five radiologists accepted in the last batch. The AI-generated report states "Normal heart size. Vascular cephalization. Pulmonary edema with bilateral pleural effusions and bibasilar consolidation or atelectasis. No acute bony abnormality." (e) An example of a radiograph and an AI-generated report that all five radiologists did not accept and modified the report in the first batch. The AI-generated report mentions "Normal cardiac silhouette. Clear lungs. No pneumonia or pneumothorax. Right-sided pleural calcifications." All five radiologists modified the AI-generated reports to include large mass the right lung zone (black arrow). (f) An example of a radiograph and an AI-generated report that all five radiologists did not accept and modified the report in the last batch. The AI-generated report describes "Rt. C-tube in situ. Decreased Rt. pneumothorax to 1.3 cm (white arrow). Normal Lt. lung. Normal heart size and pulmonary vascularity." All five radiologists modified the size of right Rt. pneumothorax and removed mention of the interval change "decreased."

Table 2. Reading time, acceptability, report agreement, and quality scores across case types (normal vs abnormal), radiologist groups, and sequential batches

	All Readers	Thoracic Radiologists	Nonthoracic Radiologists	<i>P</i> *	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5	Batch 6	Batch 7	<i>P</i> †
Reading time												
All cases	19.8 ± 24.6	10.8 ± 11.1	33.4 ± 32.0	<.001	25.8 ± 29.7	20.6 ± 21.9	19.7 ± 20.3	16.6 ± 20.3	17.8 ± 21.0	19.2 ± 24.3	19.3 ± 29.0	<.001
Normal	10.0 ± 10.2	6.2 ± 6.1	15.7 ± 12.2	<.001	14.9 ± 12.1	13.4 ± 11.7	11.5 ± 11.6	8.9 ± 9.2	9.2 ± 9.2	8.9 ± 10.8	8.9 ± 10.7	<.001
Abnormal	24.0 ± 27.5	12.8 ± 12.0	40.8 ± 34.7	<.001	42.2 ± 39.3	31.6 ± 28.2	31.9 ± 30.1	28.2 ± 26.2	30.5 ± 26.6	34.5 ± 30.2	34.9 ± 39.1	.002
<i>P</i>	<.001	<.001	<.001		<.001	<.001	<.001	<.001	<.001	<.001	<.001	
Acceptability of the AI-generated reports												
All cases	57.4 (2,171 of 3,780)	76.7 (1,739 of 2,268)	28.6 (432 of 1,512)	<.001	54.6 (295 of 540)	52.2 (282 of 540)	52.4 (283 of 540)	53.8 (315 of 540)	61.7 (333 of 540)	62.6 (338 of 540)	60.2 (325 of 540)	<.001
Normal	68.9 (768 of 1,115)	86.2 (577 of 669)	42.8 (191 of 446)	<.001	64.8 (81 of 125)	69.4 (111 of 160)	64.0 (96 of 150)	68.3 (123 of 180)	70.0 (133 of 190)	74.2 (115 of 155)	70.3 (109 of 155)	.238
Abnormal	52.6 (1,403 of 2,665)	72.7 (1,162 of 1,599)	22.6 (241 of 1,066)	<.001	51.6 (214 of 415)	45.0 (171 of 380)	48.0 (187 of 390)	53.3 (192 of 360)	57.1 (200 of 350)	57.9 (223 of 385)	56.1 (216 of 385)	<.001
<i>P</i>	<.001	<.001	<.001		.004	<.001	<.001	<.001	<.001	<.001	<.001	
Complete agreement of the final reports												
All cases	56.6 (2,139 of 3,780)	50.0 (1,134 of 2,268)	66.5 (1,005 of 1,512)	<.001	52.2 (282 of 540)	60.6 (327 of 540)	55.0 (297 of 540)	54.3 (293 of 540)	56.1 (303 of 540)	54.4 (294 of 540)	63.2 (343 of 540)	.087
Normal	57.7 (643 of 1,115)	45.6 (305 of 669)	75.8 (338 of 446)	<.001	45.6 (57 of 125)	67.5 (108 of 160)	62.7 (94 of 150)	53.9 (97 of 180)	56.3 (107 of 190)	52.9 (82 of 155)	63.2 (98 of 155)	0.128
Abnormal	56.1 (1,496 of 2,665)	51.8 (829 of 1,599)	62.6 (667 of 1,066)	<.001	54.2 (225 of 415)	57.6 (219 of 380)	52.1 (203 of 390)	54.4 (196 of 360)	56.0 (196 of 350)	55.1 (212 of 385)	63.6 (245 of 385)	.082
<i>P</i>	<.001	<.001	<.001		.102	.034	.027	.927	>.99	.703	.922	
Agreement scores												
All cases	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	<.001	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	.003
Normal	5.0 (4.5-5.0)	4.5 (4.5-5.0)	5.0 (5.0-5.0)	<.001	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	.570
Abnormal	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	<.001	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	5.0 (4.5-5.0)	.003
<i>P</i>	0.004	0.619	<.001		.391	.003	.001	.399	.516	.758	.592	
Quality scores												
All cases	4.5 (4.5-5.0)	4.5 (4.5-5.0)	5.0 (4.5-5.0)	<.001	4.5 (4.0-4.5)	4.5 (4.5-5.0)	4.5 (4.0-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	<.001
Normal	4.5 (4.5-5.0)	4.5 (4.5-5.0)	5.0 (4.5-5.0)	<.001	4.5 (4.5-4.5)	4.5 (4.5-5.0)	5.0 (4.5-5.0)	4.75 (4.5-5.0)	5.0 (4.5-5.0)	4.5 (4.5-5.0)	5.0 (4.5-5.0)	<.001
Abnormal	4.5 (4.5-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	<.001	4.5 (4.0-4.5)	4.5 (4.0-5.0)	4.5 (4.0-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	4.5 (4.5-5.0)	<.001
<i>P</i>	<.001	<.001	<.001		.002	<.001	<.001	<.001	<.001	.088	.003	

Reading times (in seconds) are presented as means ± SDs. Acceptability is shown in percentages with raw values in parenthesis. Report agreement and quality scores are presented as medians and interquartile ranges (IQRs) presented in parenthesis. *P* values were derived from the comparisons between normal and abnormal labeled cases. AI = artificial intelligence.

**P* values were derived from the comparisons between thoracic and nonthoracic radiologists.

†*P* values were derived from the regression models assessing changes over batches.

offers insights into the potential of AI to complement human expertise. It also provides brief insights into how radiologists engage with AI-generated reports, noting that although reading times decreased and acceptability increased over time, the quality of reports showed no significant difference in normal radiographs and changed without improvement in abnormal radiographs. These findings suggest that future studies should explore ways to enhance the accuracy of reports when using preliminary AI-generated reports and examine how increasing confidence in AI impacts radiologists' decision making and diagnostic performance.

Despite the increasing acceptability of AI-generated reports over time, overall diagnostic concordance of the final reports remained unchanged. This suggests that although radiologists grew more comfortable approving AI-generated reports without modifications, this did not necessarily impact the diagnostic accuracy of the final reports. Moreover, although reductions in interpretation time and increases in acceptability were statistically significant, it may not yet reflect clinically meaningful improvements in workflow or diagnostic performance. Future studies, ideally conducted prospectively in real-world clinical settings, are needed to determine whether sustained exposure to AI-generated reports alters radiologists' acceptance thresholds and how such changes may affect clinical decision making and patient outcomes.

The current study has limitations. First, including the small number of participating radiologists, which may limit the ability to assess differences across radiologists with various subspecialty training and years of experience. Second, the variability in the distribution of abnormal radiographs across batches could have influenced the results. Third, the study did not incorporate qualitative feedback from participating radiologists, and future work may benefit from interviews or surveys to better capture their perceptions and experiences when interacting with AI-generated preliminary reports. Lastly, radiologists were blinded to clinical information to control for potential confounders, however, this does not reflect real-world practice and may limit the generalizability of the findings to routine clinical settings. Future research involving a larger cohort of readers in a prospective setting is necessary to identify trends in radiologists' interaction with AI systems and to distinguish beneficial experiential adaptations, such as improved efficiency and decision making, from potential risks associated with overreliance on AI-generated outputs. Developing structured evaluation frameworks that assess and mitigate these interactions could provide valuable insights into optimizing the integration of AI in clinical workflows while maintaining diagnostic accuracy and safety.

In conclusion, this study demonstrates the potential of multimodal generative AI in improving efficiency in

radiology workflows. However, further research is needed to evaluate its impact on report quality and radiologist decision making.

TAKE-HOME POINTS

- Radiologists demonstrated a significant reduction in reading times for chest radiographs with AI-generated reports over 7 sequential batches of 108 cases each. The average reading time decreased from 25.8 ± 29.7 in batch 1 to 19.3 ± 29.0 seconds in batch 7 ($P < .001$), with normal radiographs consistently requiring less time than abnormal radiographs ($P < .001$).
- Acceptability, defined as the proportion of AI-generated reports approved without requiring revisions, increased from 54.6% in batch 1 to 60.2% in batch 7 ($P < .001$). Acceptability was higher for normal radiographs (68.9%) than for abnormal radiographs (52.6%; $P < .001$).
- Agreement scores (RADPEER criteria) and quality scores (5-point Likert scale) for normal radiographs remained stable at a median of 5.0 (IQR 4.5-5.0) and 4.5 (IQR 4.5-5.0), respectively. In contrast, abnormal radiographs displayed significant variability without a clear trend across batches in both agreement ($P = .003$) and quality scores ($P < .001$).

REFERENCES

1. Benjamins S, Dhunoo P, Meskó B. The state of artificial intelligence-based FDA-approved medical devices and algorithms: an online database. *NPJ Digit Med* 2020;3:118.
2. Çallı E, Sogancioglu E, van Ginneken B, van Leeuwen KG, Murphy K. Deep learning for chest X-ray analysis: a survey. *Med Image Anal* 2021;72:102125.
3. Huang J, Wittbrodt MT, Teague CN, et al. Efficiency and quality of generative AI-assisted radiograph reporting. *JAMA Netw Open* 2025;8:e2513921.
4. Hong EK, Ham J, Roh B, et al. Diagnostic accuracy and clinical value of a domain-specific multimodal generative AI model for chest radiograph report generation. *Radiology* 2025;314:e241476.
5. Tejani AS, Ng YS, Xi Y, Rayan JC. Understanding and mitigating bias in imaging artificial intelligence. *Radiographics* 2024;44:e230067.
6. Dratsch T, Chen X, Rezazade Mehrizi M, et al. Automation bias in mammography: the impact of artificial intelligence BI-RADS suggestions on reader performance. *Radiology* 2023;307:e222176.
7. Parasuraman R, Manzey DH. Complacency and bias in human use of automation: an attentional integration. *Hum Factors* 2010;52:381-410.
8. Bustos A, Pertusa A, Salinas J-M, De La Iglesia-Vaya M. Padchest: a large chest x-ray image dataset with multi-label annotated reports. *Med Image Anal* 2020;66:101797.
9. Hong EK, Roh B, Park B, et al. Value of using a generative AI model in chest radiography reporting: a reader study. *Radiology* 2025;314:e241646.
10. Goldberg-Stein S, Frigini LA, Long S, et al. ACR RADPEER Committee white paper with 2016 updates: revised scoring system, new classifications, self-review, and subspecialized reports. *J Am Coll Radiol* 2017;14:1080-6.